MODUPE AWOJUYIGBE

(832) 597– 0826 | awojuyigbemodupe@gmail.com | linkedin.com/in/modupe-awojuyigbe

EDUCATION

The University of Texas at San Antonio *Expected: May 2027*

- Bachelor of Science in **Computer Engineering** – GPA: **3.7/4.0**

Achievements: NSBE Scholar, Klesse Scholar, USAA Scholar, Society of Women Engineers, herScript, Rewriting the Code
Coursework: Senior Design, Computer Architecture, C++ and Data Structures, Engineering Analysis II, Microcomputer Systems

EXPERIENCE

Caterpillar Inc.
Autonomy Engineering Intern, Peoria, IL **May 2026 – August 2026**

- Led development of a multi machine autonomy proof of concept, defining functional requirements, use cases, project milestones, and technical recommendations for a new autonomous capability.
- Developed a **Python simulation** using **Pygame** and **Matplotlib** to evaluate system behavior, analyze operational scenarios, and support data driven product and engineering decisions.
- Investigated electrical and system issues through schematic review, wiring verification, component testing, and machine diagnostics, improving autonomous system performance by **20%**.

EDP Renewables
Operations Engineering Intern, Houston, TX **June 2025 – August 2025**

- Managed project scope, schedule, requirements, and deliverables for a **telemetry data-logging system**, coordinating execution across engineering teams and contractors.
- Led biweekly stakeholder meetings, tracking progress, and aligning cross-functional teams on priorities and deliverables.
- Developed **Python/Selenium** automation and data-logging solutions that reduced project setup delays by **70**, and operational costs by **$20,000**.

The University of Texas at San Antonio
Teaching Assistant / Peer Educator, San Antonio, TX **August 2024 – Present**

- Mentored **50+** students in STEM related courses, translating complex engineering concepts into real-world applications, which improved student pass rates by **60%**.

TECHNICAL PROJECTS

Personal AI Agent Assistant **Present**

- Developing an **AI powered personal assistant** using **Python, Gemini API, AWS, and API integrations**, with modular tools designed to automate email, task management, and professional workflows to improve daily productivity and efficiency.

NSBE E Board Dashboard **Fall 2026**

- Developed an E Board dashboard using **JavaScript, HTML, CSS, and Firebase**, integrating authentication and role-based access to automate participation tracking and data management.

Robotic Arm Project **Spring 2026**

- Developed an **Arduino** based robotic systems in **C++,** integrating ultrasonic sensors, infrared sensors, and servo motors for autonomous object detection, pickup, and path following.

SKILLS

- **Programming Languages:** Python, C, C++, JavaScript, HTML, CSS
- **Embedded Systems & Hardware:** Microcontrollers, Sensor Integration, Servo Motor Control, 3D Printing
- **AI & Automation:** Large Language Models (LLMs), Gemini API, AI Agents, API Integration, Git, GitHub
- **Software, Data and, Validation Tools:** Pygame, Matplotlib, Pandas, Linux, Ubuntu, Firebase, SCADA Systems, Test Plan Review, Machine Data Validation, Software Flashing, Product Status Reports
- **Professional Strengths:** Cross-Functional Collaboration, Adaptability, Problem Solving, Attention to Detail

LEADERSHIP

National Society of Black Engineers (NSBE) – *Vice President* **May 2026 – Present**

- Led executive operations for **16 officers**, tracking chapter goals, action items, and accountability while coordinating execution across the executive board; improved task follow-through by **25%**.

The University of Texas at San Antonio – *Resident Assistant* **May 2024 – August 2024**

- Supervised a community of **150+** residents while planning orientation events, improving safety compliance across **5** residence halls, and increasing first year student satisfaction by **30%.**